

ZeroDrift MK2: shopping list

Team ZeroDrift · Jabalpur Engineering College · SIH 2026 · for Amar Robotics

For the shopkeeper: we are building a small pan-tilt camera turret with two **NEMA17 stepper motors (5 mm shaft)** and **A4988 drivers**, run by an Arduino Nano. We need the items below.

Must buy

NO.	ITEM (ASK FOR)	QTY	DESCRIPTION FOR THE SHOPKEEPER	≈ ₹
☐ 1	5 mm rigid flange coupling	2	Metal flange coupling with a 5 mm bore , to fit a NEMA17 stepper motor shaft. Rigid (solid) type, not the flexible or spring type. Must have a grub screw and a flange with 4 screw holes. Sometimes sold as a "5 mm wheel hub / shaft coupler".	160–300
☐ 2	12 V 2 A DC adapter (SMPS)	1	Wall adapter, output 12 V DC, 2 A , with a 5.5 × 2.1 mm barrel plug. It powers the two stepper motors. (Our 5 V adapter is too weak for motors.)	200–350
☐ 3	DC female jack with screw terminals	1	Female barrel jack, 5.5 × 2.1 mm, matching item 2, with two screw terminals marked + and -. Lets us connect the adapter to wires without cutting it.	20–40
☐ 4	Resistor 100 Ω, ¼ W	5	Carbon film resistor, 100 ohm, quarter watt. Colour bands brown-black-brown-gold . For the decoy LED.	5–10
☐ 5	Resistor 220 Ω, ¼ W	5	Carbon film resistor, 220 ohm, quarter watt. Colour bands red-red-brown-gold . For the beacon LED.	5–10
☐ 6	Red LED, high brightness	2	Bright red LED, 10 mm (5 mm also fine), clear or red lens. One for the decoy, one spare.	10–20
Total for the must-buy items				≈ ₹400–730

Only if needed

NO.	ITEM (ASK FOR)	QTY	DESCRIPTION · BUY IT ONLY WHEN...	≈ ₹
☐ 7	TP4056 charging module (with protection)	1	Single-cell 18650 Li-ion charging board, USB-C or micro-USB. Only if the shop has no 12 V adapter (item 2): we then charge our 18650 cells one at a time and run the motors from them.	30–60

☐	8	Red cellophane or red gel sheet	1	A small sheet of transparent red film, to go over the camera lens. Only if there is no red film or old sunglasses lens at home. Stationery shops have it.	10–30
☐	9	USB cable for Arduino Nano	1	Mini-USB or USB-C, the same as the second Nano's port. Only if we don't already have the right cable.	60–120
☐	10	USB-C hub (USB-C to 2× USB-A)	1	Only if the laptop has USB-C ports only: the Nano and the webcam must be plugged in at the same time.	300–800

Already have: don't buy

- ✓ 2× NEMA17 stepper motors
- ✓ 2× NEMA17 L-brackets
- ✓ 2× AS5600 encoders + diametric magnets
- ✓ 2× Arduino Nano
- ✓ Red and white LEDs
- ✓ Ping-pong ball
- ✓ 5 V 1 A adapter (for the beacon)
- ✓ Brass standoffs, M3 screws and nuts
- ✓ Multimeter, soldering iron, glue gun
- ✓ USB webcam (arriving)
- ✓ 2× A4988 drivers (with heatsinks)
- ✓ 100 µF capacitors
- ✓ TCA9548A multiplexer
- ✓ KY-008 laser
- ✓ 3×AA battery holder
- ✓ 3× 18650 cells + holder, B3 charger
- ✓ Toggle switches
- ✓ Breadboard + jumper wires
- ✓ White sheet (base plate)
- ✓ Foam tape, tape, scissors

Find at home: no need to buy

- △ 2 small plastic boxes (tiffin, soap box): beacon and decoy
- △ Short brass standoffs screwed together to ≈ 40 mm: head post
- △ Superglue (Fevikwik): magnets
- △ Old CD or stiff lid: the two small discs
- △ Old sunglasses lens: camera red filter
- △ Drill, or someone who has one (about 10 holes)

Remember: the B3 charger never connects to the rig; it only charges batteries. The 5 V 1 A adapter can't run the motors (it's for the beacon). The motors run on 12 V: item 2, or charged 18650s (3.7–4.2 V each).